

Decision Tree Regression – Complete Step-by-Step Worked Example

Dataset Used:

We predict Salary using two features: Experience (years) and Education Level (1=UG, 2=Masters, 3=PhD).

Experience | Education | Salary

1 | 1 | 30

2 | 1 | 35

3 | 2 | 60

4 | 3 | 80

5 | 3 | 85

6 | 3 | 90

Step 1: Compute Initial Mean and MSE

Mean Salary = $(30+35+60+80+85+90) / 6 = 63.33$

Initial MSE = $[(30-63.33)^2 + (35-63.33)^2 + (60-63.33)^2 + (80-63.33)^2 + (85-63.33)^2 + (90-63.33)^2] / 6$

= $(1111.11 + 802.78 + 11.11 + 277.78 + 469.44 + 711.11) / 6$

= $3383.33 / 6$

= 563.89

Step 2: Try Split on Education (Threshold = 2)

Left Node (Education ≤ 2): 30, 35, 60

Mean Left = 41.67

Left MSE = $[(30-41.67)^2 + (35-41.67)^2 + (60-41.67)^2] / 3 = 172.22$

Right Node (Education > 2): 80, 85, 90

Mean Right = 85

Right MSE = 16.67

Weighted MSE = $(3/6)*172.22 + (3/6)*16.67 = 94.44$

Variance Reduction = $563.89 - 94.44 = 469.45$

Step 3: Try Split on Experience (Threshold = 3.5)

Left Node ($\text{Experience} \leq 3.5$): 30, 35, 60

Right Node: 80, 85, 90

This produces the same grouping as Education split.

So Weighted MSE = 94.44

Variance Reduction = 469.45

Root split can be either Education or Experience (equal gain).

Step 4: Split Left Node Further

Current Left Node Data: 30, 35, 60

Current MSE = 172.22

Try $\text{Experience} \leq 2$:

Left: 30, 35 $\rightarrow$ Mean = 32.5 $\rightarrow$ MSE = 6.25

Right: 60 $\rightarrow$ MSE = 0

Weighted MSE = $(2/3) * 6.25 = 4.17$

Variance Reduction = $172.22 - 4.17 = 168.05$

This is a strong split, so we choose Experience at this node.

Step 5: Final Tree Structure

Root: $\text{Education} \leq 2$

Left: $\text{Experience} \leq 2$

Predict 32.5 (for 30,35)

Predict 60

Right:

Predict 85 (mean of 80,85,90)

Each leaf predicts the mean of values inside it.

Important Concepts Demonstrated

1. Tree tries ALL features at every node.
2. It calculates weighted MSE for each possible threshold.
3. It selects the split with maximum variance reduction.
4. Features can be reused multiple times.
5. Feature importance equals total variance reduction contributed by that feature.
6. Trees create piecewise constant prediction regions.
7. Overfitting happens if we keep splitting until pure leaves.